

MTH209 Midsem Part A, IIT Kanpur

27 Feb 2026 | Time : 20 minutes | Total Points : 20

System Instructions

- Please do not open the quiz in multiple tabs/browsers. All your responses may not be recorded.
- Responses are automatically saved.

Quiz Instructions

- This part has 10 questions. 2 marks per question.
- Focus checker is on. So you are not allowed to leave the quiz window.

Q1. UMAP and tSNE are both dimension reduction techniques

✓ **True**

- False

Q2. Mean squared error is the only way to assess the quality of an estimator.

- True

✓ **False**

Q3. Linear regression is embarrassingly parallel.

- True

✓ **False**

Q4. Which of the following is embarrassingly parallelizable?

- median

✓ **mean**

✓ **max**

- mode

Q5. The tSNE method gives a different solution every time we run it on a dataset because

- the data keeps changing
- the perplexity is randomly chosen

✓ **the optimization problem is non-convex**

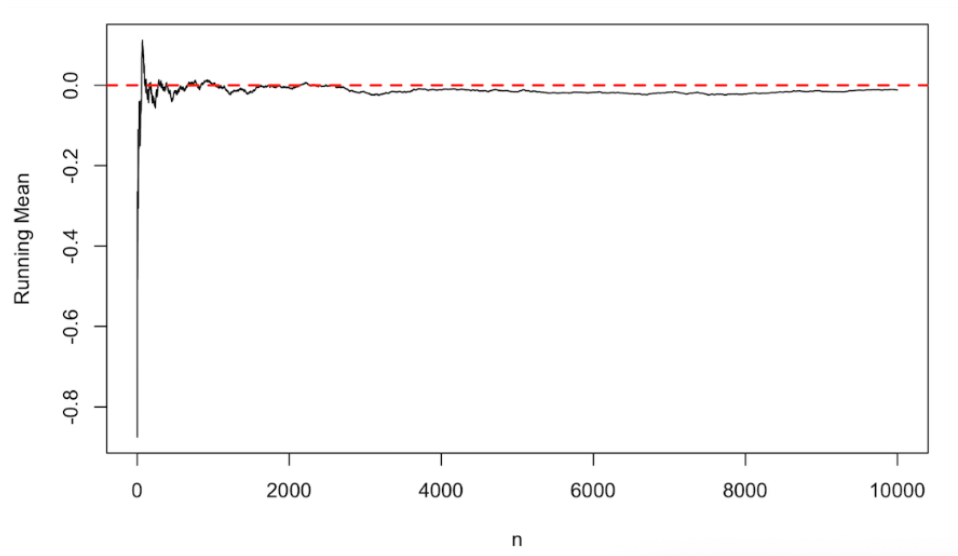
- the optimization problem is concave

Q6. The singular value decomposition can be used to invert any square symmetric positive-definite matrix.

✓ **True**

- False

Q7. What statistical property is the plot below verifying?



- Unbiasedness
- Biasedness
- ✓ **Consistency**
- Central limit theorem

Q8. Let $X \sim N_{10}(0, \Sigma)$, where Σ is a 10×10 positive-definite matrix. Which of the following is true for $Y = AX$ for a 10×10 matrix A .

- $\text{Var}(Y) = A^T \Sigma A$
- If A has negative eigenvalues, then $\text{Var}(Y)$ is not positive definite.
- ✓ **$E(Y) = 0$**
- $\text{Var}(Y) = \mathbb{I}_{10}$ when $A = \Sigma^{-1}$

Q9. Singular value decomposition of a matrix of the data observations yields the principal components.

- True
- ✓ **False**

Q10. Let $A \in \mathbb{R}^{n \times p}$ and $B \in \mathbb{R}^{p \times m}$ be two matrices. Consider parallelizing the product AB over p cores. What is the computational complexity per core?

- $O(nmp)$
- ✓ **$O(nm)$**
- $O(np)$
- $O(mp)$